DIGITAL SIGNAL ENCODER AND DECODER

Technical Specification Report

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1. Objective and Scope

The project implements digital bit-stream generation, line encoding, AMI scrambling, longest-palindromic-substring detection, and analog-to-digital conversion using Pulse Code Modulation (PCM). A Raylib graphical interface visualizes encoded signals, analog samples, and the PCM staircase.

2. Implemented Features

- **Digital generation:** Random bits, streams containing runs of four/eight zeros where possible, and manual bit entry.
- **Line coding:** NRZ-L, NRZ-I, Manchester, Differential Manchester, and AMI.
- **Scrambling:** Optional B8ZS or HDB3 processing for AMI.
- **PCM:** Configurable sinusoidal amplitude, frequency, sample count, and bits/sample; PCM encoding/decoding and staircase visualization. PCM bits can be passed to a line encoder.
- **Palindrome:** Manacher's algorithm finds the longest contiguous palindromic substring in $O(n)$ time.

3. Assumptions and Conversion Conventions

- **Input:** Digital input contains only `0` and `1`. Manual input is limited to 128 bits. The longest palindrome is a contiguous substring, not a subsequence.
- **NRZ-L:** Bit `0` maps to $+1$ (high level); bit `1` maps to -1 (low level).
- **NRZ-I:** Bit `1` causes a transition at the start of its interval; bit `0` causes no transition. The initial previous level is $+1$.
- **Manchester:** Bit `0` maps to $(-1, +1)$ (low-to-high); bit `1` maps to $(+1, -1)$ (high-to-low).
- **Differential Manchester:** Every bit has a mid-bit transition. Bit `0` causes a transition at the start; bit `1` does not. The initial reference level is $+1$.
- **AMI:** Bit `0` maps to 0. Successive `1`s alternate between $+1$ and -1; the first `1` maps to $+1$.
- **B8ZS:** Eight consecutive zeros are replaced by the pattern $000VB0VB$, relative to the preceding pulse polarity. The normalized pattern in the implementation is $(0, 0, 0, +1, -1, 0, -1, +1)$, multiplied by the preceding pulse polarity. The last pulse polarity is unchanged.
- **HDB3:** Four consecutive zeros are replaced by $B00V$ for even pulse-count parity and $000V$ for odd parity, according to the implementation. In $B00V$, B has polarity opposite to the preceding pulse and V has the same polarity as B. In $000V$, V has the polarity of the preceding pulse, creating a bipolar violation.
- **PCM:** A sinusoidal input is sampled and quantized into fixed-width binary codewords. Amplitude and frequency must be positive; sample count is 1–128 and resolution is 1–16 bits/sample. Total PCM output is limited to 128 bits.

4. Software Requirements

Language	C++17
Graphics	Raylib
Build tools	GCC, MSYS2 UCRT64, and `pkg-config`
Resource	Keep `times.ttf` in the project root alongside the executable; launch the program from that directory.

5. Build and Execution

Prerequisites: Install GCC, Raylib, and `pkg-config` in the MSYS2 UCRT64 environment.

Step 1 – Compile: Open the MSYS2 UCRT64 terminal in the project root directory and execute:

```
g++ -std=c++17 -Wall -Wextra -Wpedantic \
src/gui.cpp src/encoders.cpp src/input_engine.cpp \
src/palindrome.cpp $(pkg-config --cflags --libs raylib) \
-o gui.exe
```

Step 2 – Run: Launch the compiled application using:

```
./gui.exe
```

6. Testing and Verification

Five test executables passed, covering 38 tests in total: basic encoders, B8ZS/HDB3 cases, input generation and validation, palindrome detection, and PCM decoding. Manual GUI checks included amplitude/frequency changes, PCM output limits, and staircase display.

7. Acknowledgment

The project was developed through my own understanding of digital signal encoding principles and hands-on implementation. OpenAI ChatGPT was used as a supporting resource during development to investigate programming errors, refine selected implementation details, and identify relevant test cases. I evaluated suggestions in the context of my code, made appropriate changes, and tested the application to verify its behaviour. The tool complemented my development process rather than replacing my own understanding and involvement in the project.